

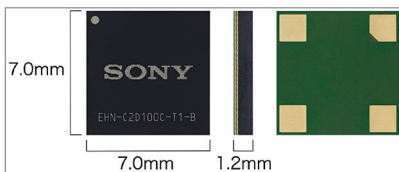
TRULY INNOVATIVE ELECTRONICS

INNOVATION UPDATES

Amongst numerous press releases of new products received by us, these are the ones we found worthy of the title *Truly Innovative Electronics*

Electromagnetic energy harvester

Sony Semiconductor Solutions Corporation (SSS) has launched an energy harvesting module that uses electromagnetic wave noise energy. Such technology can tap into the consistent electromagnetic disturbances produced



by factory robots, office equipment like monitors and lights, and household and commercial devices such as TVs and monitors. This can ensure a reliable power source for low-energy IoT devices and communication tools. The innovative approach involves harnessing the electromagnetic wave noise ubiquitously produced by electronic equipment. Despite its compactness, it can transform electromagnetic wave disturbances ranging from several Hz to 100MHz into electrical energy. This can then power low-energy IoT sensors and communication devices or recharge batteries. This pioneering approach marks the industry's first energy harvesting method, offering high power generation efficiency.

Sony Semiconductor Solutions Corporation
<https://www.sony-semicon.com>

Microcontroller with advanced I3C features and versatile design

The PIC18-Q20 microcontroller series of Microchip is a feature-rich chip available in 14- and 20-pin configurations and comes with multi-voltage I/O support. It represents the market's

first low-pin-count microcontroller equipped with an improved inter-integrated circuit (I3C) target mode. The microcontroller stands out as the premier general-purpose microcontroller with a low pin count to integrate this new interface. The PIC18-Q20

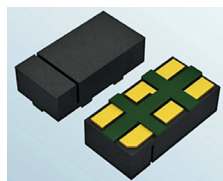


has enhanced I3C features, including in-band interrupts, dynamic address provisioning, and hot-join functionalities. The unique features include its core logic's ability to function at a voltage lower than the I3C interface.

Microchip
<https://www.microchip.com>

Compact proximity sensor for diverse applications

ROHM has introduced the RPR-0720, a compact 2mm x 1mm proximity sensor designed for proximity detection and attachment/detachment needs. The



RPR-0720 utilises a vertical cavity surface emitting laser (VCSEL) for its light emission and a dedicated IC sensor for light reception. By using specialised chips, this module reduces its size by approximately 78% compared to conventional products. The applications include discerning the attachment or detachment of devices like wireless earbuds, smartwatches, gaming console controllers, and VR headsets. Additionally, it can spot the presence of paper in office equipment such as printers, copiers, and shredders. The system

is also efficient in identifying cartridge or liquid presence in e-cigarettes and monitoring various conditions in smartphones and laptops.

ROHM
<https://www.rohm.com>

Revolutionising AI with first of its kind server on a chip

NeuReality's advanced 7nm NR1 AI-focused chip is the world's first AI-centric server on a chip (SoC). Its applications include natural language processing, computer vision, and



recommendation systems. In evaluations with AI-focused server systems, the NR1 chip showcased a tenfold performance at equivalent costs to traditional CPU-focused systems. NeuReality is reshaping the industry by facilitating a shift from a predominantly software-driven method to a hardware-enhanced approach. The consistent hardware acceleration in the company's automated SDK flow simplifies the process for organisations of all sizes, ensuring they benefit from top-tier performance, minimised power usage, and cost-effective infrastructure. This also provides user-friendly operations for AI inferencing services.

NeuReality
<https://neureality.ai>

A processor with record core count for cloud innovation

Ampere Computing has launched the AmpereOne series of processors, featur-